

ITD Assignment-2

- 1) Employee (ename, ecity, state)
WORKS (ename, company name, salary)
Company (company name, city)
Manages (ename, managername)

(a)

Relational algebra

$$\pi_{ename} (\sigma_{salary > 60000 \wedge (WORKS)} \\ (companyname = 'TCS' \cup \\ companyname = 'WIPRO'))$$

Tuple Relational Calculus

$$\{t.ename \mid WORKS(t) \wedge t.salary > 60000 \wedge (t.companyname \\ = 'TCS' \cup t.companyname = 'WIPRO')\}$$

Domain Relational Calculus

$$\{e \mid \exists e, s (WORKS(e, e, s) \wedge s > 60000 \wedge (e = 'TCS' \cup \\ e = 'WIPRO'))\}$$

(b) Relational algebra

$$\pi_{ename, salary} (WORKS \bowtie_{WORKS.companyname = Company.companyname} \\ \sigma_{city = 'DELHI'} (Company))$$

TRC

$$\{(t.ename, t.salary) \mid \text{WORKS}(t) \wedge \exists c (\text{Company}(c) \wedge \\ c.companyname = t.companyname \\ \wedge c.city = 'DELHI')\}$$

DRC

$$\{(e, s) \mid \exists c (\text{WORKS}(e, c, s) \wedge \text{Company}(c, 'DELHI'))\}$$

(c)

Relational algebra

$$\pi_{salary}(\sigma_{city='MUMBAI' \wedge \\ managename='JOHN'}(\text{Employee} \bowtie \text{Manages} \bowtie \text{WORKS}))$$

TRC

$$\{w.salary \mid \text{WORKS}(w) \wedge \text{Employee}(e) \wedge \text{Manages}(m) \wedge \\ e.ename = w.ename \wedge m.ename = w.ename \\ \wedge e.city = 'MUMBAI' \wedge m.managename \\ = 'JOHN'\}$$

DRC

$$\{s \mid \exists e, c, m (\text{Employee}(e, 'MUMBAI', m) \wedge \text{WORKS}(e, c, s) \\ \wedge \text{Manages}(e, 'JOHN'))\}$$

2> Student (SID, Name, City, Age)
 Course (CID, CName, Dept, Credits)
 Enroll (SID, CID, Marks)
 Teacher (TID, TName, Dept, Salary)
 Teaches (TID, CID)

(a) $\pi_{TName} (\sigma_{Dept = 'CSE' \wedge Salary < 80000 \wedge CName = 'ITD'} (Teacher \bowtie Teaches \bowtie Course))$

(b) $\pi_{Name, Marks, Credits} (Student \bowtie Enroll \bowtie Course)$

(c) $\pi_{CID} (\sigma_{T1.TID \neq T2.TID} (\rho_{T1}(Teaches) \bowtie_{T1.CID = T2.CID} \rho_{T2}(Teaches)))$

3> Employee (EId, Name)
 Brand (BId, BName)
 Own (EId, BId)

Relational
 algebra

$\pi_{EId} (Own \div \pi_{BId} (Brand))$

TRC

$$\{e. EID \mid \text{Employee}(e) \wedge \forall b (\text{Brand}(b) \rightarrow \exists o (\text{Own}(o) \wedge o. EID = e. EID \wedge o. BID = b. BID))\}$$

DRC

$$\{e \mid \forall b (\text{Brand}(b) \rightarrow \exists o (\text{Own}(e, b)))\}$$